

Objective classification of extratropical cyclogenesis

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Extratropical cyclones experience vastly different genesis conditions at the first point of their tracks. A novel method is introduced to characterize this variability and classify genesis events by computing 30 diagnostic variables that describe the synoptic-scale environment of 16 029 genesis events in the Northern Hemisphere extratropics, using ERA-Interim reanalyses from 2000–2011. These variables are referred to as precursors and include parameters characterizing upper-level forcing, low-level baroclinicity, thermodynamic stability, surface fluxes and moist processes. The genesis events spread over a large portion of the 30-dimensional precursor phase space and no obvious clusters occur, which highlights the high variability of cyclogenesis processes and indicates that they form in a continuum rather than a few distinct categories. A projection of the genesis events to the first two principle components (PC) of the precursor phase space allows reduction of the dimensionality and introduction of a meaningful segmentation of the genesis events in five classes. The first two PCs are characterized by upper-level forcing (e.g. the amplitude of the upper-level potential vorticity (PV) anomaly) and low-tropospheric diabatic processes (e.g. precipitation and diabatically produced low-level PV), respectively. The first of the five classes identified constitutes the centre of the PC1–PC2 phase space and represents average conditions. Composites reveal that the four classes of events characterized by large positive or negative scores of PC1 and PC2 occur in distinct and strongly differing flow regimes, characterized by the strength of the upper-level forcing, the structure of the upper-level jet and the amplitude of low-level moist processes and baroclinicity. The four classes also have clearly differing geographical distributions. Many well-known cyclogenesis events fall within classes characterized by strong low-level moist processes with or without strong upper-level forcing. Also discussed are the robustness of the method and the linkage to classical concepts of cyclone classifications.

Key Words: extratropical cyclogenesis; objective classification; principal component analysis; potential vorticity

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1. Introduction

Adverse weather in the extratropics is in most cases related to the passage of an extratropical cyclone with its accompanying fronts. Early weather analyses, based upon synoptic charts of sea-level pressure (SLP), revealed the transient and propagating nature of extratropical cyclones and motivated research on the genesis, track and intensity of these important synoptic-scale atmospheric flow features. Very early examples of such cyclone case studies are from Loomis (1841) and Shaw (1903). Since then, many further studies have shown that extratropical cyclones are highly variable in terms of e.g. their lifetime and propagation speed (e.g. Blender *et al.*, 1997), maximum intensity (e.g. Hodges *et al.*, 2011), size (e.g. Simmonds, 2000) and their three-dimensional structure (e.g. Campa and Wernli, 2012). This high degree of variability calls for a reduction of complexity by building either so-called conceptual models or

meaningful categorizations of extratropical cyclones. Bjerknes and Solberg (1922) proposed a conceptual life-cycle model for a prototype extratropical cyclone, which very successfully influenced meteorological education and set the standard for drawing surface weather charts. This model emphasized the key role of a pre-existing surface front for cyclogenesis. At about the same time, Ficker (1920) pointed out the important role of upper-tropospheric signatures and their superposition with near-surface flow features for cyclogenesis (see also Davies, 2010). These two streams of thinking, focusing on either the lower- or upper-level ingredients of developing cyclones, evolved further in parallel and constitute the conceptual backbone for much of the extratropical cyclone research performed during the last century (Davies, 1997). Today it is generally accepted that the variability of observed cyclone life cycles is so large that they cannot be described with a single conceptual model. This understanding prompts the question as to whether archetypical categories of

cyclones exist and, if so, how they can be determined. Catto (2016) recently summarized many different attempts to categorize cyclones based on theoretical and conceptual models on the one side and observational data sets, e.g. satellite imagery, on the other side.

A first approach is to focus on cyclones in specific regions, assuming that they form a distinct cyclone category due to the geographical setting, for instance Mediterranean cyclones (e.g. Trigo *et al.*, 1999) and Icelandic Lows (e.g. Serreze *et al.*, 1997), or to compare characteristics of cyclones developing in different regions (e.g. Hoskins and Hodges, 2002; Dacre and Gray, 2009; Campa and Wernli, 2012). An alternative classification is based upon cyclone intensity or the degree of cyclone intensification, measured for instance by the temporal change of SLP along the cyclone track (Sanders and Gyakum, 1980; Binder *et al.*, 2016). A third approach, which is more comprehensive and less straightforward, is to classify extratropical cyclones according to the environmental flow conditions at the time of their genesis and/or intensification. In their classical article, Petterssen and Smebye (1971) investigated cyclogenesis events and postulated two contrasting types of cyclone development, referred to as types A and B, respectively. Whereas cyclones of type A develop 'under a more or less straight upper current', type B cyclones occur when a 'pre-existing upper trough, with strong vorticity advection on its forward side, spreads over a low-level area of warm advection'. The latter type clearly emphasizes the key role of finite-amplitude upper-level precursors for cyclogenesis, which, from a potential vorticity (PV) perspective, correspond to intense upper-tropospheric positive anomalies (Kleinschmidt, 1950; Hoskins *et al.*, 1985; Campa and Wernli, 2012; Bentley *et al.*, 2016). Many case studies of cyclogenesis have confirmed the key role of this type of upper-level PV precursor (e.g. Bleck and Mattocks, 1984; Tafferner, 1990; Morgan and Nielsen-Gammon, 1998; Rossa *et al.*, 2000; Lagouvardos *et al.*, 2007; Iwabe and da Rocha, 2009) and idealized numerical model studies on baroclinic instability have used this 'type B paradigm' to trigger baroclinic instability (e.g. Montgomery and Farrell, 1992; Schär and Wernli, 1993; Wernli *et al.*, 1998; Schemm *et al.*, 2013). In contrast, type A cyclones develop first at low levels without a clear upper-level disturbance at the time of genesis. Such a set-up is typical for so-called diabatic Rossby waves (e.g. Wernli *et al.*, 2002; Moore *et al.*, 2008; Rivière *et al.*, 2010; Boettcher and Wernli, 2011), but it is not clear whether Petterssen and Smebye (1971) thought mainly of this relatively rare type of cyclone when describing type A. The twofold categorization of cyclone development by Petterssen and Smebye (1971) has subsequently been extended by Deveson *et al.* (2002) to include a third category of cyclones, type C, the dynamics of which is dominated by the effect of intense latent heating and the resulting diabatic production of positive PV anomalies in the lower to middle troposphere (see also Plant *et al.*, 2003; Gray and Dacre, 2006; Dacre and Gray, 2009). However, for cyclones of type C also, the genesis process is strongly determined by an upper-level PV anomaly (Plant *et al.*, 2003). This threefold categorization highlights the importance of upper-level disturbances, surface baroclinicity and diabatic processes in extratropical cyclogenesis.

The currently available classifications of extratropical cyclones are useful for highlighting the strongly differing dynamics leading to cyclone formation and intensification. However, they might not encompass the full diversity of cyclogenesis events observed in nature and, due to their focus on a few pre-selected parameters, do not necessarily account for all physical processes involved. For instance, detailed case studies emphasized the important precursory role of surface latent heat fluxes for intense oceanic cyclones (e.g. Kuo *et al.*, 1991; Gyakum and Danielson, 2000; Davolio *et al.*, 2009). In contrast, for cyclones in the lee of high mountains, cold-air blocking and interaction with the upper-level trough (e.g. Egger *et al.*, 1995; Han *et al.*, 1995; Horvath *et al.*, 2006) and adiabatic warming of surface air due to downslope flows are of key importance (e.g. Thomas and Martin, 2007).

Other documented case studies extend the palette of potentially important precursors of cyclogenesis considerably. These include, for instance, intense surface frontogenesis for the explosive 'Queen Elizabeth II' storm (Gyakum, 1991), the compacting of an upper-level PV disturbance for another rapidly intensifying storm over the North Atlantic (Lackmann *et al.*, 1997), warm sea-surface temperature and weak static stability for subtropical cyclogenesis (Evans and Guishard, 2009), low-level anomalies of equivalent potential temperature for cyclones over South America (Mendes *et al.*, 2007) and the presence of an intense upper-level jet streak in the upstream quadrant in the incipient stage of intense cyclones near Australia (Sinclair and Cong, 1992). For frontal wave cyclones, several studies have investigated the mechanism proposed by Bishop and Thorpe (1994), who suggested that environmental deformation must be below a certain threshold for frontal waves to grow (e.g. Renfrew *et al.*, 1997; Rivals *et al.*, 1998; Chaboureaud and Thorpe, 1999; Mallet *et al.*, 1999; Schemm and Sprenger, 2015).

This brief summary of selected earlier studies clearly indicates that we still lack a comprehensive investigation of the relative importance of various physical processes and precursor signals for the genesis of extratropical cyclones in different regions of the globe. It is therefore the objective of this study to perform such an investigation, leading to a novel and objective classification of cyclogenesis events. Our method focuses on cyclogenesis, considered as the starting point of a cyclone track, uses a cyclone-track climatology based upon reanalysis data, computes a large set of potential cyclogenesis precursors in the vicinity of the genesis events and applies objective statistical techniques to identify meaningful categories of cyclogenesis. The methodological steps are introduced in the next section. Sections 3 and 4 then introduce and characterize the genesis categories and section 5 provides a discussion of the robustness of the results, the linkage to earlier categorizations and the relevance for individual cyclone case studies. The main conclusions are summarized in section 6.

2. Cyclogenesis events and their precursors

2.1. Identification of cyclogenesis events

Locations of cyclogenesis are identified in the present study as the initial points of cyclone tracks, which are identified in the Northern Hemisphere during 2000–2011 according to the method described in Wernli and Schrierz (2006). This method relies on sea-level pressure (SLP) fields from ERA-Interim (Dee *et al.*, 2011), interpolated on a regular grid with 1° horizontal resolution, and the three following steps: (i) SLP contours are identified at 0.5 hPa intervals; (ii) local SLP minima are determined on the regular grid and accepted for further analysis if they are surrounded by at least one closed isobar; (iii) the accepted SLP minima are then tracked and kept if the track duration is at least 24 h. The initial point (longitude, latitude, time) of each cyclone track finally defines a cyclogenesis event.

In addition to this basic tracking methodology, several subtleties have to be considered. First, only cyclogenesis events in the extratropics (i.e. north of 30°N) are considered, in order to exclude tropical systems. Furthermore, to avoid artificial SLP minima due to extrapolation in regions of high topography, all minima are neglected where topography is higher than 1500 m. Then, if the cyclone area at the genesis point, defined as the area within the outermost enclosing contour, is larger than $2 \times 10^5 \text{ km}^2$, the point is neglected, because we assume that such events are caused by incorrectly interrupted cyclone tracks and an already mature (i.e. large) cyclone is hence erroneously captured as a new system. Finally, if several local minima are located within the same enclosing contour, only the cyclogenesis event with the highest core pressure is kept.

Using this method, we identify 16 934 cyclogenesis events in the Northern Hemisphere extratropics during 2000–2011. As a caveat we mention that Neu *et al.* (2013) showed that different

Table 1. List of precursors with abbreviation (**Abbrev.**), associated signal (**Signal**) and statistical values of the normalized climatological anomalies (**Std**, **Mean**, **Skew**). In column **Signal**, the (+/−) sign means that in the cyclogenesis area the average above/below the median is calculated (see text for details).

Short description of precursor	Abbrev.	Signal	Std	Mean	Skew
Upper-level PV (averaged between 600 and 200 hPa)	PV_{up}	+	0.93 (5)	0.73	0.60
Geopotential height at 500 hPa	Z_{500}	−	0.90	−0.63	−0.03
Anomaly of geopotential height at 500 hPa	Z_{ANOM}	−	0.94 (4)	−0.78	−0.66
Low-level contribution to QG vertical motion	$qg\omega_{bot}$	−	0.85	− 1.34 (2)	−1.21
Upper-level contribution to QG vertical motion	$qg\omega_{top}$	−	1.07 (3)	− 1.37 (1)	−1.24
Wind speed averaged between 500 and 100 hPa	$VELJET$	+	0.88	0.57	0.44
Tropospheric static stability	N_{TROPO}^2	−	0.72	−0.82	0.11
Difference of pot. temperature between surface and 700 hPa	$\Delta\theta_{SFC700}$	−	0.74	−0.66	0.24
Mixed-layer CAPE	$CAPE$	+	1.08 (2)	0.45	4.42 (2)
Eady growth rate in lower troposphere	$EADY$	+	0.83	0.94	0.36
Eady growth rate in upper troposphere	$EADY_{up}$	+	0.78	0.64	0.51
Horizontal temperature gradient at 850 hPa	$ \nabla\theta_{850} $	+	0.81	1.08 (5)	0.55
Horizontal temperature advection at 850 hPa	T_{ADV850}	+	0.64	0.86	1.32 (5)
Equivalent potential temperature at 850 hPa	θ_{e850}	+	0.76	0.86	−0.13
Horizontal gradient of equivalent pot. temperature at 850 hPa	$ \nabla\theta_{e850} $	+	0.76	1.03	0.62
Environmental deformation at 850 hPa	DEF	+	0.84	1.09 (4)	0.86
Petterssen frontogenesis function at 850 hPa	$FGEN_{850}$	+	0.87	1.02	1.86 (3)
Anomaly of the 850 hPa temperature	T_{PERT}	+	0.81	0.88	0.13
Low-level PV (averaged between 1000 and 600 hPa)	PV_{low}	+	0.93 (5)	0.99	0.19
Vertically integrated water vapour	Q_{int}	+	0.92	0.98	0.37
Surface precipitation during 6 h before cyclogenesis	RR_{6h}	+	1.53 (1)	1.33 (3)	1.64 (4)
Time period since moisture uptake	$UPTTIM$	+	0.52	0.66	−0.12
Difference of pot. temperature in 48 h back. trajectories	$TRAd\theta_{48}$	+	0.63	0.92	0.71
Pressure difference in 48 h back. trajectories	$TRAdP_{48}$	−	0.46	−0.99	0.95
Maximum change of pot. temperature in 48 h back. trajectories	$TRAd\theta_{MX}$	+	0.53	0.99	0.39
Maximum pressure difference in 48 h back. trajectories	$TRAdP_{MX}$	−	0.55	−1.06	0.43
Sensible heat flux at surface	$SSHf$	+	0.66	0.60	−0.17
Latent heat flux at surface	$SLHF$	+	0.58	0.62	−0.23
Skin temperature	SKT	+	0.78	0.54	−4.68 (1)

Std shows the standard deviation, **Mean** the arithmetic mean and **Skew** the skewness of the normalized climatological anomalies. For reasons of comparability, none of the precursor distributions is log-transformed for the calculation of statistical values. The top five highest absolute values are indicated with bold letters and the ranking is shown in brackets.

tracking algorithms generally agree well during the intensification and mature phases of a cyclone's life cycle, but differences exist during the early cyclogenesis phase. Our cyclone identification and tracking approach tends to identify cyclones later than some other schemes (which, for example, are based on vorticity maxima at 850 hPa: Hodges, 1999), due to the requirement that closed SLP contours already exist at the time of cyclogenesis.

2.2. Precursor definition and normalization

In order to classify the 16 934 cyclogenesis events, a set of 30 diagnostic variables that describe the synoptic-scale environment surrounding each cyclone are calculated from the ERA-Interim dataset. These variables are referred to as precursors; they are listed in Table 1 and described in greater detail in the Supplementary Material. The precursors were selected after an in-depth literature review on case studies of extratropical cyclones. The chosen precursors were mentioned in at least one study as relevant for the genesis of the event. They include dynamical flow features at upper and lower levels (e.g. upper and low-level PV), parameters related to baroclinicity, fronts and static stability and other thermodynamic parameters.

All 30 precursors have been calculated for all 16 934 cyclogenesis events. With regard to this technical procedure, some aspects deserve a more detailed discussion.

- What is a reasonable approach to represent a two-dimensional field in the environment of a genesis event (e.g. upper-level PV) by a scalar value?
- Is it more meaningful to consider actual fields or deviations from climatology (i.e. anomalies)?

- How shall the precursors be normalized for an objective classification?
- How should we treat fields with a dipolar structure?

In response to question (i) above, an area average within a broad region surrounding each cyclone is used to represent the synoptic-scale environment associated with each cyclogenesis event. An example of this is shown for cyclone 'Klaus', located at 44°N, 29°W at 0600 UTC on 23 January 2009. Figure 1 shows the upper-level PV (PV_{up}) within a $20^\circ \times 20^\circ$ region surrounding the location of cyclogenesis. Only a weak PV_{up} signal occurs exactly at the genesis position; however, a distinct trough, i.e. a positive PV_{up} signal, is located to the west. To account for this, all precursor fields are considered within a circle of radius 500 km around the genesis position (from now on referred to as the cyclogenesis area; see dashed circle in Figure 1).

In response to questions (ii) and (iii) above, we decided to use normalized anomaly fields. More precisely, anomaly fields were calculated as the difference between the actual fields and climatology, which we calculated at a given time t as a 30 day running average from $t - 15$ days to $t + 15$ days in the 12 years considered (2000–2011). This anomaly field is then normalized by the standard deviation of the climatology (i.e. the standard deviation is calculated from the 12 yearly running mean values). This procedure is shown in Figure 1 for cyclone Klaus: panel (a) shows the climatological distribution of PV_{up} (black contours) and panel (b) the normalized and dimensionless PV_{up} anomaly. The trough to the west of the genesis event, which is believed to play an important role, appears as a strong positive anomaly of PV_{up} , with an amplitude of almost 3 standard deviations of the interannual variability. Because we normalize with local standard deviations (instead of uniform spatially averaged

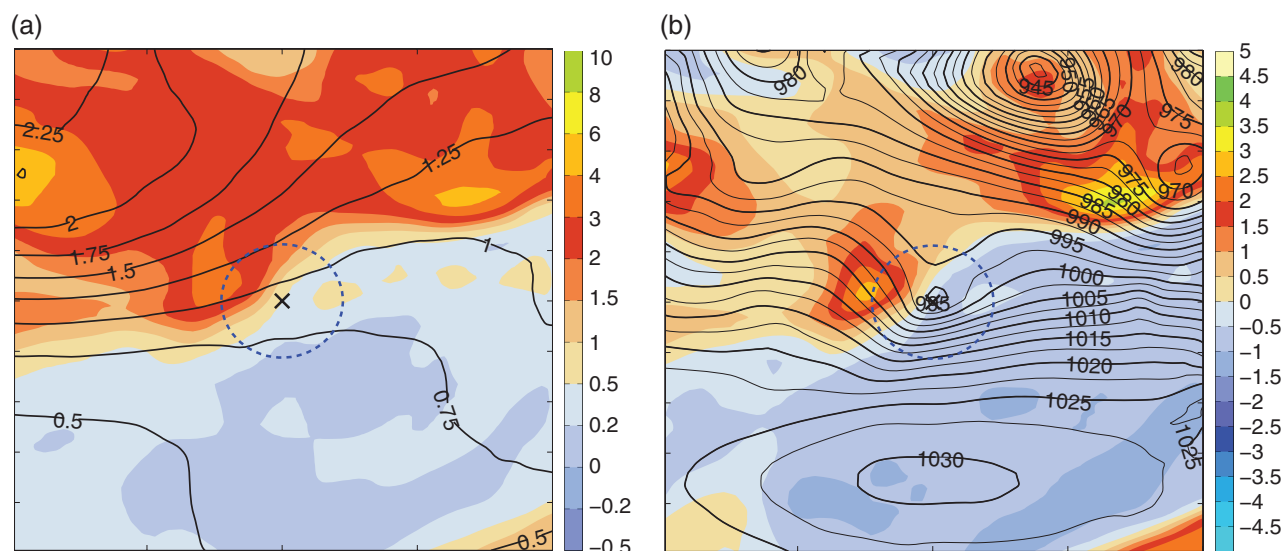


Figure 1. Cyclogenesis of extratropical cyclone *Klaus* at 44°N, 29°W at 0600 UTC on 23 January 2009. The cyclogenesis is located in the centre of the domain (marked with a black cross); the horizontal domain extends 2000 km in each direction (corresponding to 20° at the Equator). (a) Upper-level PV at cyclogenesis (in puv) is colour-shaded and the corresponding climatology shown in black contours. (b) Normalized upper-level PV anomaly is colour-shaded, where normalization is relative to a 30 day running mean (see text for details), and sea-level pressure (in hPa) is shown as black contours. In both panels, the blue circle defines the cyclogenesis area around the cyclogenesis point.

standard deviation), this approach attributes more weight to parameters that exhibit particularly strong variability near genesis events. Some precursors have a strongly skewed distribution: surface precipitation during 6 h before cyclogenesis (RR_{6h}), the Petterssen frontogenesis function at 850 hPa ($FGEN_{850}$) and mixed-layer CAPE ($CAPE$); they are log-transformed after the normalization to obtain a more symmetric normal-like distribution.

Figure 1(b) illustrates the challenge associated with question (iv) above. The trough–ridge pattern in the genesis region leads to a dipolar structure with strongly positive values to the west and negative values to the east. However, considering the type B paradigm (see introduction) we are interested in the amplitude of the upper-level positive PV anomaly and therefore not in a simple spatial average of the normalized PV anomaly field in the cyclogenesis area. We therefore decided to average only the positive or negative anomaly values in the genesis area for each parameter. More precisely, the method applied in this study is first to consider the precursor values at all grid points in the cyclogenesis area and to determine their median value. For each field, the average over all grid points either above ($med+$) or below ($med-$) the median is chosen as the final precursor signal. To decide which of the two options is more reasonable, we considered the statistics of the ($med+$) and ($med-$) values. As an example, Figure 2 shows the distribution and mean of $PV_{up}(med+)$ (red) and $PV_{up}(med-)$ (blue) for all cyclogenesis events. The former distribution deviates more strongly from zero and therefore $PV_{up}(med+)$ is chosen as the parameter representing the PV_{up} precursor. In Table 1, the choice of the ($med+$) values is marked with a + symbol. For some parameters (see Table 1), ($med-$) is the more appropriate choice, e.g. for the geopotential height at 500 hPa (Z_{500}), which makes sense given the expected anti-correlation of PV_{up} and Z_{500} .

At least 20 grid points of each precursor field have to be computable within the cyclogenesis area, otherwise the whole cyclogenesis event is removed from the list. Non-computable grid points can occur, for instance, if the calculation of quasi-geostrophic (QG) forcing for vertical motion does not converge numerically, or if the considered field, e.g. temperature at 850 hPa, is missing because of topography. Due to these reasons, 905 events could not be considered for the classification, resulting in a precursor dataset for 16 029 cyclogenesis events.

Finally, note that the precursors do not all describe well-separated physical processes: some parameters correlate or can be attributed to similar forcing mechanisms. This aspect of

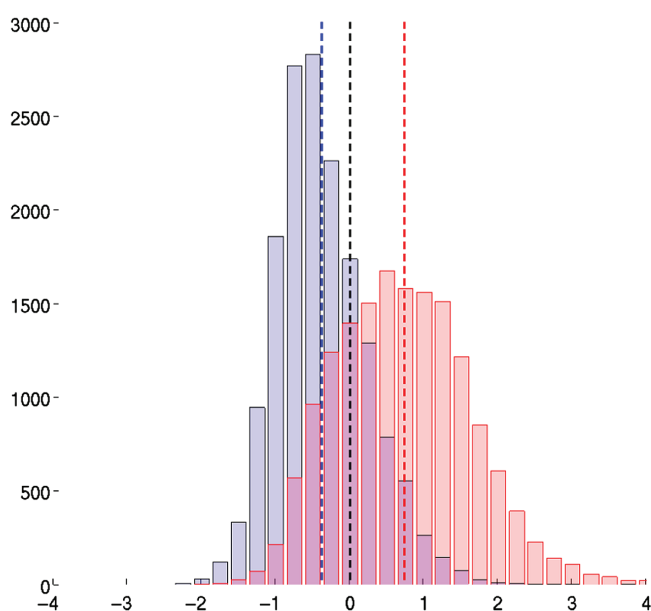


Figure 2. Distribution of $PV_{up}(med-)$ (blue) and $PV_{up}(med+)$ (red) for all cyclogenesis events. The blue (red) dashed lines show the mean of the two distributions. The signal $PV_{up}(med-)$ is calculated as the mean of all PV_{up} values in the cyclogenesis area below the median and correspondingly for $PV_{up}(med+)$ (see text for details). The black dashed line marks the position of zero anomaly.

the precursors is handled reasonably well by the classification approach described in section 3.

2.3. Statistical distribution of precursors

Here, some relevant characteristics of the precursor distributions are considered: (i) the general form of the distribution, e.g. their skewness; (ii) the spread of the distribution; and (iii) the shift of the precursor signal from its climatological mean. Remember that we normalized the precursors with respect to a climatological distribution and therefore it is interesting to determine whether genesis events show substantial deviations from this climatology. An objective classification scheme will favour any precursor that exhibits a considerable spread over all cyclogenesis events. In this sense, these basic statistics of the precursor distributions already point to their potential relevance in the objective classification.

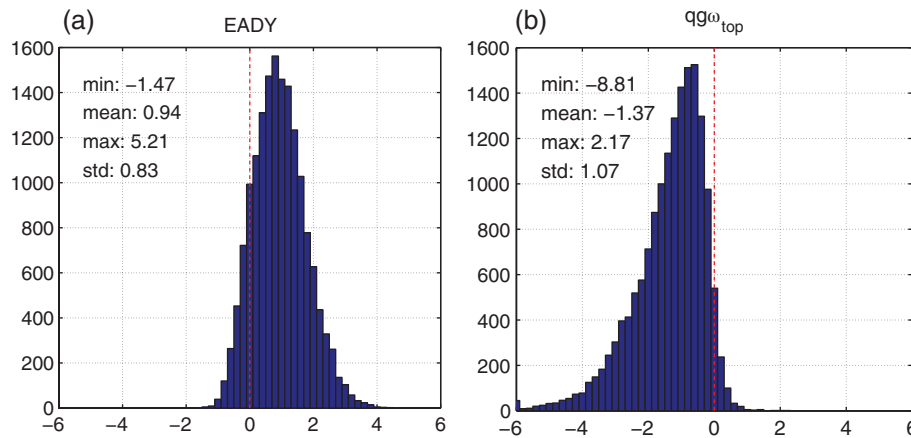


Figure 3. Statistical distribution of the normalized anomalies for a selection of precursors: (a) Eady growth rate, defined as $EADY(\text{med}+)$; (b) upper-level contribution to QG vertical motion, defined as $qg\omega_{\text{top}}(\text{med}-)$. Additionally, in each panel the following values are given: minimum, maximum, mean and standard deviation.

Most precursors show a unimodal normal-like statistical distribution (two examples are shown in Figure 3). The Eady growth rate in the lower troposphere ($EADY$) has a rather small skewness of 0.36 (Table 1), which reflects the normal-like shape of the distribution with a slightly longer tail for positive values. The mean of $EADY$ is positive (0.94), which indicates that genesis typically occurs where and when the Eady growth rate is anomalously strong, as physically expected. However, it must be kept in mind that our method to define the precursor signal introduces a slight bias, because it is defined as $EADY(\text{med}+)$ (see section 2.2). Finally, the standard deviation (0.83) indicates that, quantitatively, this signal can differ strongly from event to event and that some genesis events occur where and when this parameter is below climatology.

As a second example, Figure 3(b) shows the upper-level contribution to QG vertical motion ($qg\omega_{\text{top}}$), with pronounced negative skewness (-1.24), a mean negative value (-1.37) and a significant standard deviation (1.07). Again, the mainly negative values of this precursor are influenced by our choice to consider the ($\text{med}-$) values (see Table 1). What can be learned from the distribution is that, in almost all cases, a region with forcing for upward motion is found in the cyclogenesis area and this upward forcing can be strong, with values reaching six climatological standard deviations. As for $EADY$, the considerable spread of $qg\omega_{\text{top}}$ (standard deviation of 1.07) indicates that this parameter might be of importance for the later classification.

Table 1 also lists the statistical measures for all other precursors. It is worth highlighting the following general aspect. If a precursor has a very small standard deviation, this implies that the (normalized) precursor signal is very similar for all genesis events. This is the case, e.g., for the surface heat fluxes ($SSHf$ and $SLHF$) and the change of potential temperature along 48 h backward trajectories ($TRAd_{48}$; three-dimensional kinematic trajectories have been calculated with the tool introduced by Wernli and Davies, 1997). These precursors can be relevant for cyclogenesis; however, the objective classification introduced in section 3.1 by design focuses on precursors that are relevant for the *variability* of genesis events. It is also insightful to compare the mean and standard deviation values. If the mean is clearly larger than the standard deviation, the precursor signal nearly always has the same sign. As an example, the horizontal temperature advection at 850 hPa (T_{ADV850}) has a mean of 0.86 and a standard deviation of 0.64, i.e. genesis goes along with enhanced temperature advection compared with climatology in (almost) all cases. On the other hand, PV_{up} can be higher or lower than the corresponding climatological value, because the standard deviation is larger (0.93) than the mean (0.73). Some further interesting conclusions from Table 1 are as follows: (i) particularly strong deviations from the mean are found for the upper-level and low-level contributions to QG vertical motion

($qg\omega_{\text{top}}$ and $qg\omega_{\text{bot}}$, respectively), the horizontal temperature gradient at 850 hPa ($|\nabla\theta_{850}|$), the environmental deformation at 850 hPa (DEF) and RR_{6h} ; (ii) $CAPE$ and skin temperature (SKT) exhibit a particularly strong skewness, followed by intermediate skewness for T_{ADV850} , $FGEN_{850}$ and RR_{6h} ; and (iii) the largest spread is found for RR_{6h} , $CAPE$, $qg\omega_{\text{top}}$, the anomaly of geopotential height at 500 hPa (Z_{ANOM}) and PV_{up} .

3. The biplot and definition of cyclogenesis classes

The aim of this section is twofold. First, the first two principal components (PC) of the 30-dimensional precursor phase space are used to define five distinct classes objectively. Secondly, a physical interpretation is sought for the projection of the precursors on these two PCs.

3.1. Principal component analysis (PCA)

A PCA is performed (Pearson, 1901; Hotelling, 1933), based on a singular value decomposition. Each genesis event is represented as a point in a 30-dimensional phase space, where the 30 dimensions correspond to the (normalized) precursors. Accordingly, the 16 029 genesis events constitute a point cloud in this phase space. The PCA redefines the coordinate axes in such a way that most of the variance of the point cloud is found along the first PCA axis (PC1). The second PCA axis (PC2) is then chosen such that it maximizes variance in the subspace orthogonal to PC1, and so on for the other 28 PCA axes. These redefined PCA axes can be expressed as linear combinations of the original precursors, which allows the PCs to be interpreted in terms of the original precursors. Formally, the projection of a genesis point onto the i th PC, the so-called score s^i , can be written as follows:

$$s^i = \sum_{j=1}^{N_p} \lambda_j^i \cdot Y_j. \quad (1)$$

Here, N_p denotes the dimension of the precursor space (30) and Y is a vector containing the N_p precursor values for one genesis event. λ_j^i is the so-called loading of the j th precursor in the direction of the i th PC. An advantage of such a PCA approach is that it enables reduction of the discussion to only the first few PCs, in particular if they explain a large fraction of the variance in the data set.

In this study, we consider the projection of all cyclogenesis events on to the first two PCA axes, i.e. s^1 (PC1) and s^2 (PC2), which corresponds to a reduction of the 30-dimensional initial precursor space to a simpler two-dimensional space. Of course,

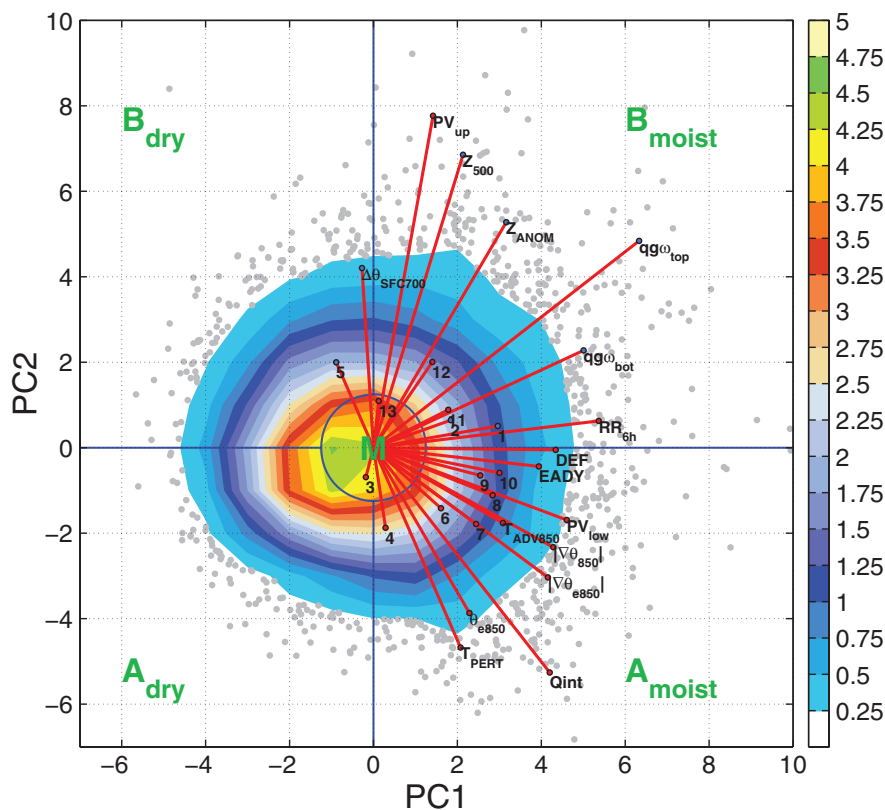


Figure 4. The density of cyclogenesis events (multiplied by 100) in the principal component space is colour-shaded. Each grey dot corresponds to an outlier cyclogenesis event, with the PC1 and PC2 scores as coordinates. The red lines represent the coefficients (loadings) of the precursors, allowing the new PC1 and PC2 axes to be interpreted in terms of the original precursors (see text). If the coefficients point towards positive (negative) anomalies, they have a red (blue) dot at the end. The most important precursors are labelled with abbreviations defined in Table 1. The others are labelled with numbers: 1 = $TRADP_{MX}$, 2 = $TRADP_{48}$, 3 = $UPTTIM$, 4 = SKT , 5 = N_{TROPO}^2 , 6 = $SLHF$, 7 = $SSHf$, 8 = $EADY_{up}$, 9 = $VELJET$ 10 = $FGEN_{850}$, 11 = $TRAD\theta_{MX}$, 12 = $TRAD\theta_{48}$, 13 = $CAPE$.

the reduced data set only captures part of the variability of all genesis events. Here, PC1 and PC2 explain 22.3% and 18.9% of the total variance, respectively. PC3 and PC4 are significantly less important, with an explained variance of 10.2% and 8.5%, respectively. Because of this gap in explained variance between the first two and the other PCs, it is meaningful to consider only the first two. This reduction to two dimensions facilitates the classification and further characterization of the classes considerably.

Figure 4 is the key plot of our PCA. The horizontal axes correspond to PC1 and PC2, respectively, the colour shading describes the density of cyclogenesis events in PC1 and PC2 space and the grey dots represent outlier genesis events (according to their PC1 and PC2 scores). Each precursor is shown by a radial vector (in red). Its direction indicates the direction of maximum correlation with the first two PCs and the squared length of the vectors represents approximately the variance of the corresponding precursor. This visualization is referred to as a biplot (Gabriel, 1971; Jolliffe, 2002) and interpreted in the next paragraph. First, however, we note that no obvious clusters are discernible in the density plot of genesis events, which would have been facilitating the definition of classes. Therefore, it is to be accepted that the classes we will define are not characterized by well-defined borders. Physically, this indicates that nature provides a rather continuous spectrum of cyclogenesis events. The blue lines in Figure 4 denote the five classes. We decided to select first a ‘middle class’ *M*, which contains the 20% ($n = 3206$ in absolute numbers) of all events that are closest to the cloud mean. The other four classes correspond to the four quadrants outside class *M*. They are referred to as classes *A_{dry}* ($n = 3509$), *A_{moist}* ($n = 3244$), *B_{dry}* ($n = 3160$) and *B_{moist}* ($n = 2910$). As will be discussed below, the *A* and *B* classes share similarities with the classical categorization by Petterssen and Smebye (1971) and the moist and dry classes are characterized by strong and weak moist processes, respectively.

3.2. Physical interpretation of the biplot

We now consider the red precursor vectors in the biplot (Figure 4). Long vectors contribute strongly to PC1 and PC2 and the relationship between two precursors is given by the scalar product of their vectors. Orthogonal vectors are essentially linearly independent and (anti-)parallel vectors are strongly (anti-)correlated. Note that all these inferences depend on the explained variance of the first two PCs. The higher it is, the more accurate the given interpretation of the precursor vectors (Gabriel, 1971, 1972).

Important precursors in the direction of PC1 are RR_{6h} , PV_{low} , $|V\theta_{850}|$, DEF , the horizontal gradient of the equivalent potential temperature at 850 hPa ($|V\theta_{e850}|$) and $EADY$. They are all related to either latent heat release or low-level baroclinicity, indicating that high values of PC1 are strongly associated with latent heat release along an intense low-level baroclinic zone. Important precursors in the direction of PC2 are PV_{up} , Z_{500} and Z_{ANOM} , i.e. precursors directly related to the strength of cyclonic upper-level flow anomalies. This fairly clear distinction between low-level and upper-level precursors is a promising result of our analysis and not something that could be expected a priori. Hence, the variability between cyclogenesis events is determined most strongly by the varying strength of low-level baroclinicity and latent heating on the one hand and upper-level disturbances on the other hand. Interestingly, the vectors for QG forcing for ascent, $qg\omega_{top}$ and $qg\omega_{bot}$, are rather long and point in a direction between the axes PC1 and PC2, i.e. they are about equally correlated with parameters of baroclinicity and upper-level disturbances (which theoretically would be expected). Also, the fact that $qg\omega_{top}$ aligns more with PC2 and $qg\omega_{bot}$ with PC1 agrees well with the fact that PC1 is more strongly determined by low-level and PC2 by upper-level precursors.

Another remarkable result is that PV_{up} and PV_{low} are nearly orthogonal to each other, which means that they are essentially

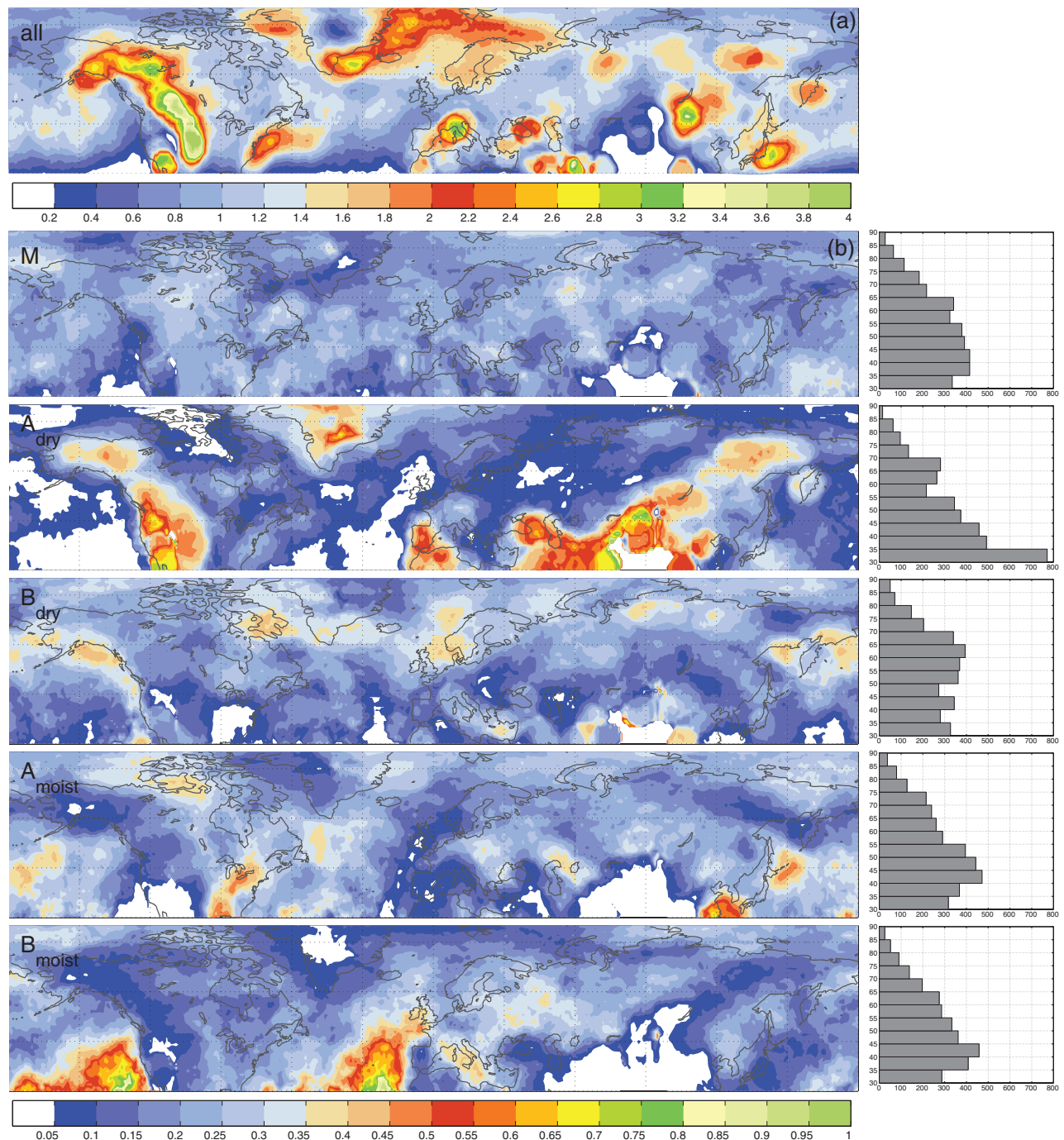


Figure 5. (a) Density map of cyclogenesis events during 2000–2011 (values in colours correspond to the number of cyclones within a circle of 500 km radius). (b) Fraction of cyclogenesis density in the different classes (in colours). The sum over all classes is one. The histograms on the right show the number of cyclogenesis cases in different latitude bins.

statistically uncorrelated and can be regarded as independent elements of cyclogenesis. Of course, other precursor pairs are strongly correlated: e.g., as expected, Z_{500} , Z_{ANOM} and PV_{up} . As a final example, consider the precursor of the tropospheric static stability (N_{TROPO}^2 ; label 5 in Figure 4): It is physically plausible that it is almost parallel to the other stability precursor that represents the difference of potential temperature between the surface and 700 hPa (Δ_{SFC700}), because both precursors are related to static stability, even if Δ_{SFC700} only considers the lower troposphere. They both point in the direction of PV_{up} , indicating that a strong upper-level PV anomaly goes along with a substantial reduction of static stability beneath (note that PV_{up} has a ‘+’ signal, whereas both stability measures have a ‘−’ signal; see Table 1).

3.3. Geographical distribution

In this subsection, we consider the geographical density of all cyclogenesis events and their distribution among the five classes

(Figure 5). Preferred locations of cyclogenesis are found, e.g. downstream of the Rocky Mountains and Greenland, between the Alps and the northern part of the Mediterranean and in the North Atlantic and North Pacific storm track regions, in particular their western parts.

Class M does not show particularly preferred regions of cyclogenesis compared with the overall climatology. This is expected, because M cyclones are non-extreme in all characteristics, by definition. In contrast, the frequency of class A_{dry} cyclogenesis is considerably reduced in the oceanic basins, reflecting that latent heat release is not an important precursor in this class. A_{dry} cyclones are more frequent over mountainous terrain, e.g. over the Rocky Mountains, Greenland and in Central Asia, indicating that many class A_{dry} events likely represent cases of lee cyclogenesis. In class B_{dry} , cyclogenesis occurs further north compared with A_{dry} . This shift becomes particularly apparent over the continents. B_{dry} cyclones are relatively frequent in some oceanic regions, e.g. south of Alaska, near the southern

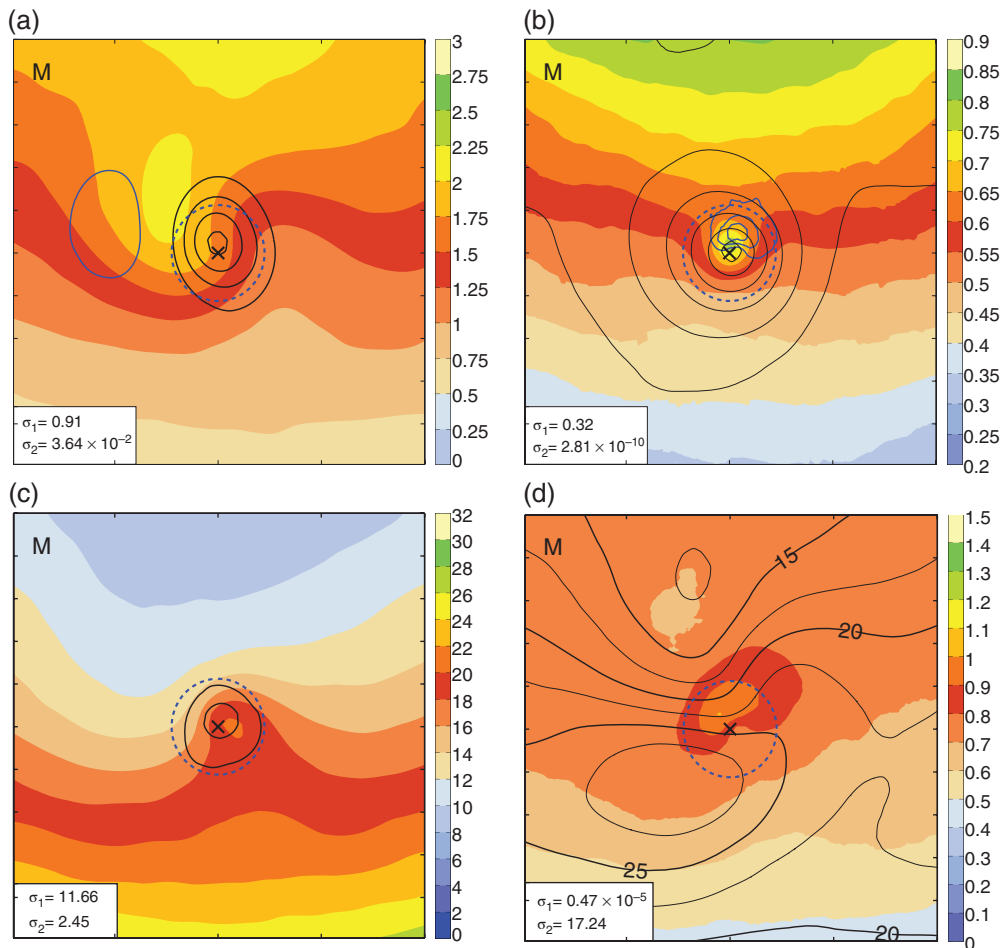


Figure 6. Composites of the middle class. The black cross marks the position of the cyclogenesis point and the blue dashed ring defines the cyclogenesis area. (a) Upper-level PV (PV_{up} , in pvu) is colour-shaded and the black contours represent upper-level induced QG vertical motion ($qg\omega_{top}$) in steps of $10^{-2} \text{ Pa s}^{-1}$. Blue and black lines indicate ascending and descending motion, respectively. (b) Low-level PV (PV_{low} , in pvu) is colour-shaded and Petterssen's frontogenesis function ($FGEN_{850}$) is represented as blue solid lines in steps of $0.5 \times 10^{-10} \text{ K m}^{-1} \text{ s}^{-1}$, beginning with $1 \times 10^{-10} \text{ K m}^{-1} \text{ s}^{-1}$. The thin black line shows sea-level pressure in 2 hPa steps. (c) Vertical integrated water vapour (Q_{int} , in mm) is colour-shaded and the black contours represent the six-hourly total surface precipitation (RR_{6h}) in steps of 1 mm. (d) Eady growth rate ($EADY$) is colour-shaded in 10^{-5} s^{-1} and the black contours show the horizontal wind speed averaged over range 100–500 hPa ($VELJET$) in m s^{-1} . σ_1/σ_2 are the standard deviations of the colour-shaded/contour values at exactly the cyclogenesis location.

tip of Greenland and in the eastern Mediterranean. For class A_{moist} , the North Atlantic and North Pacific storm tracks stand out. In orographic regions, e.g. in the Mediterranean or over the Rocky mountains, the frequency of class A_{moist} cyclogenesis is reduced. Class A_{moist} cyclogenesis is more prevalent in the western North Atlantic than e.g. class B_{dry} cyclones and is frequent over the eastern US and around the Yellow Sea. Finally, class B_{moist} cyclogenesis occurs considerably more often in the southeastern parts of the North Atlantic and North Pacific and in the western Mediterranean. In these regions, anticyclonic Rossby-wave breaking and the formation of upper-level PV cut-offs is frequent (e.g. Wernli and Sprenger, 2007), which is in agreement with the strongly positive upper-level PV anomalies in this category.

4. Characterization of classes

In the previous section, five classes of cyclogenesis were defined in terms of the first two principal components of the 30-dimensional precursor phase space. A first physical interpretation of the different classes was given by means of the biplot (see Figure 4). In particular, it was shown that the classes clearly differ with respect to the amplitude of the upper-level precursors, as well as low-level moist processes and baroclinicity. In this section, the class characteristics are analyzed further with composites of the spatial structure of the precursor fields. We start with a general characterization of the main precursors fields for the middle class and then proceed with a comparison of the main precursors for the four edge classes. The composites of the less important

precursor fields not discussed in this section can be found in File S1.

4.1. Characteristics of the middle class

By definition, class M is characterized by low absolute values of both PC1 and PC2, which implies not particularly strong or weak values of upper-level forcing, low-level diabatic processes and baroclinicity. In this sense, we consider these events as average cyclogenesis situations. Eight precursor composites are shown in Figure 6. In all composites, the cyclogenesis event is located in the centre (marked with a cross) and the domain extends about 2000 km in the four horizontal directions. Note that the composites do not show the normalized precursor fields (as in Figure 1(b)), but rather the full fields (as in Figure 1(a)), to facilitate physical interpretation.

Figure 6(a) shows PV_{up} with a value of about 1.5 pvu near the location of cyclogenesis. In addition to the background meridional gradient of PV_{up} (reflecting the increase of the Coriolis parameter towards the pole), a distinct positive upper-level PV anomaly (or PV trough) is discernible upstream (to the left) of the cyclogenesis event. Downstream, a PV ridge prevails, leading to a significant zonal PV gradient near the location of cyclogenesis. Note that the PV trough is essentially north-south oriented. This upper-level PV structure induces mid-tropospheric and low-level vertical lifting, as confirmed by the pattern of $qg\omega_{top}$ (Figure 6(a)). On the upstream side of the PV trough, there is QG forcing for descent, while on its downstream side there is strong forcing for vertical lifting with a maximum exactly over the cyclogenesis location.

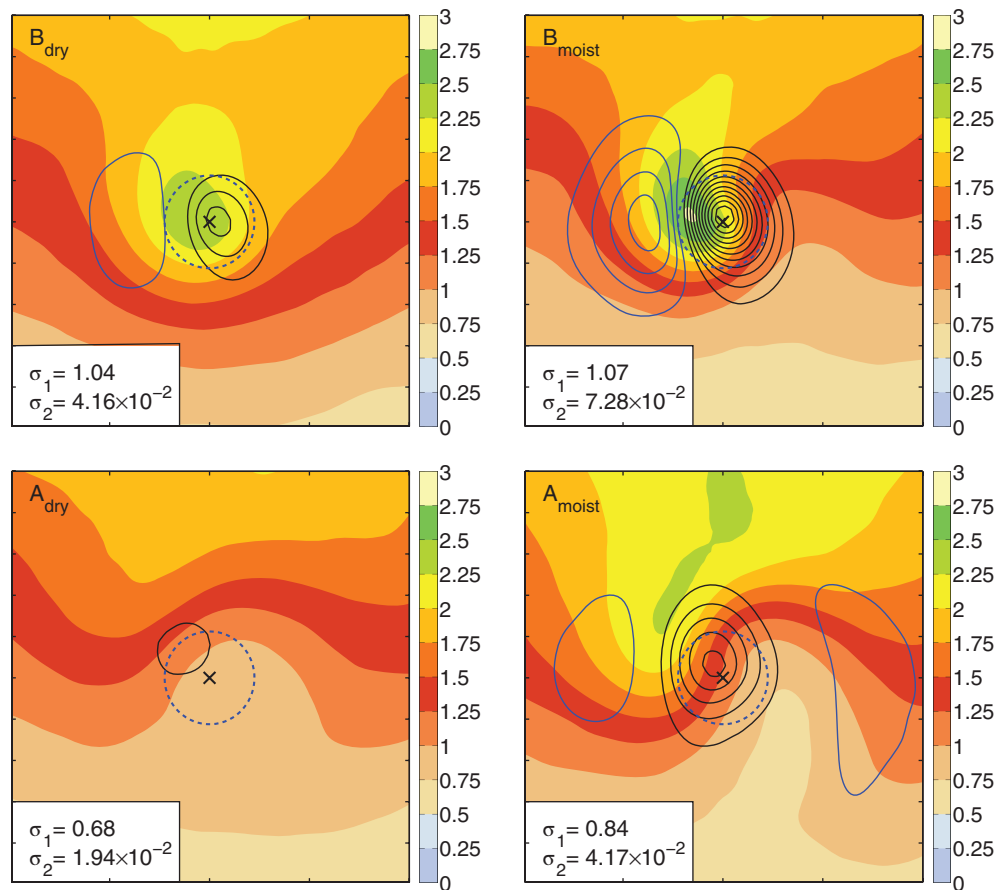


Figure 7. PV_{up} (in colour) and $qg\omega_{top}$ (contours) as in Figure 6(a), but for the edge classes A_{dry} , A_{moist} , B_{dry} and B_{moist} .

The low-level precursors PV_{low} and $FGEN_{850}$ are shown in Figure 6(b). Again, a meridional background PV gradient is discernible, of course now exhibiting typical tropospheric PV values (≈ 0.4 – 0.7 pvu). The PV gradient is zonally rather unperturbed, except for the region very close to cyclogenesis, where PV attains values that are twice as large as in the surrounding. We interpret this distinct PV signal as diabatically produced, associated with latent heat release due to cloud condensation. $FGEN_{850}$ attains a distinct maximum slightly northeast of cyclogenesis, in the region where a warm front is likely to form. Note that it is not clear whether this frontogenesis is a prerequisite of cyclogenesis or rather an effect of the cyclonic circulation acting on the background baroclinicity. In the same region, low-level warm air advection (not shown) is also enhanced. Again, this can directly impact the cyclone but also be signal of the cyclonic circulation itself.

In Figure 6(c) vertically integrated water vapour Q_{int} and surface precipitation in the 6 h before cyclogenesis RR_{6h} are shown. Also Q_{int} features a clear meridional gradient corresponding to the background temperature decrease towards the pole. Interesting for cyclogenesis is the distortion of the Q_{int} contours near the center: A tongue of enhanced Q_{int} extends from the south over the cyclogenesis region and the highest values are found to the east of cyclogenesis. For RR_{6h} a local maximum occurs near the cyclogenesis location, which coincides well with the enhanced low-level PV (Figure 6(b)). This suggests the diabatic origin of the low-level PV maximum shown via PV_{low} .

Finally, Figure 6(d) depicts the Eady growth rate in the lower troposphere ($EADY$). An area of enhanced values is found immediately north of the cyclogenesis location. This implies that on average cyclogenesis does not occur where $EADY$ has a local maximum but further south. Additionally, $VELJET$ reveals an upper-level jet streak ($> 30 \text{ m s}^{-1}$) southwest of the cyclogenesis location. Accordingly, cyclogenesis in the middle class takes place in the left exit region of this jet streak, i.e., in a region that is well known as favorable for cyclogenesis.

4.2. Characteristics of edge classes

We now consider composite fields for the four edge classes: A_{dry} , B_{dry} , A_{moist} , and B_{moist} . Composites of the most important precursors will be shown for each of these four classes, inter-compared, as well as compared to class M . The arrangement of panels in Figures 7–10 is such that it corresponds to the quadrants of the biplot (Figure 4). A concise summary for each class will be provided in the conclusions. Note again that the precursor composites not discussed in this section can be found in File S1.

4.2.1. Upper-level PV and QG forcing for ascent

Figure 7 shows composites of PV_{up} and $qg\omega_{top}$ for all edge classes (see Figure 6(a) for class M). Only a weak PV signal is discernible for class A_{dry} , where cyclogenesis is located to the east of a weak PV ridge or even within a weak PV trough. This PV structure goes along with nearly no upper-level forcing for lifting. Compared with the composite of class M (Figure 6(a)), the upper-level disturbance in class A_{dry} is much weaker. Of course, these results are in full agreement with the biplot (Figure 4), because the vectors for PV_{up} and $qg\omega_{top}$ are pointing away from the quadrant of class A_{dry} .

The class B_{dry} is located beneath a distinct cyclonic anomaly of PV_{up} , which strongly resembles a PV cut-off. Its maximum is found slightly to the west of the cyclogenesis and it is embedded in a rather broad PV trough. This strong PV_{up} signal is in perfect agreement with the biplot in Figure 4, where the PV_{up} vector points into the B half-plane. This is also the case for the $qg\omega_{top}$ signal, which is fairly moderate, although a dipole in vertical forcing is associated with the PV maximum. Cyclogenesis in class B_{dry} coincides fairly well with the maximum forcing for ascent. Figure 4 shows that N_{TROPO}^2 is relatively small (not shown), which can be expected beneath a strong positive upper-level PV anomaly.

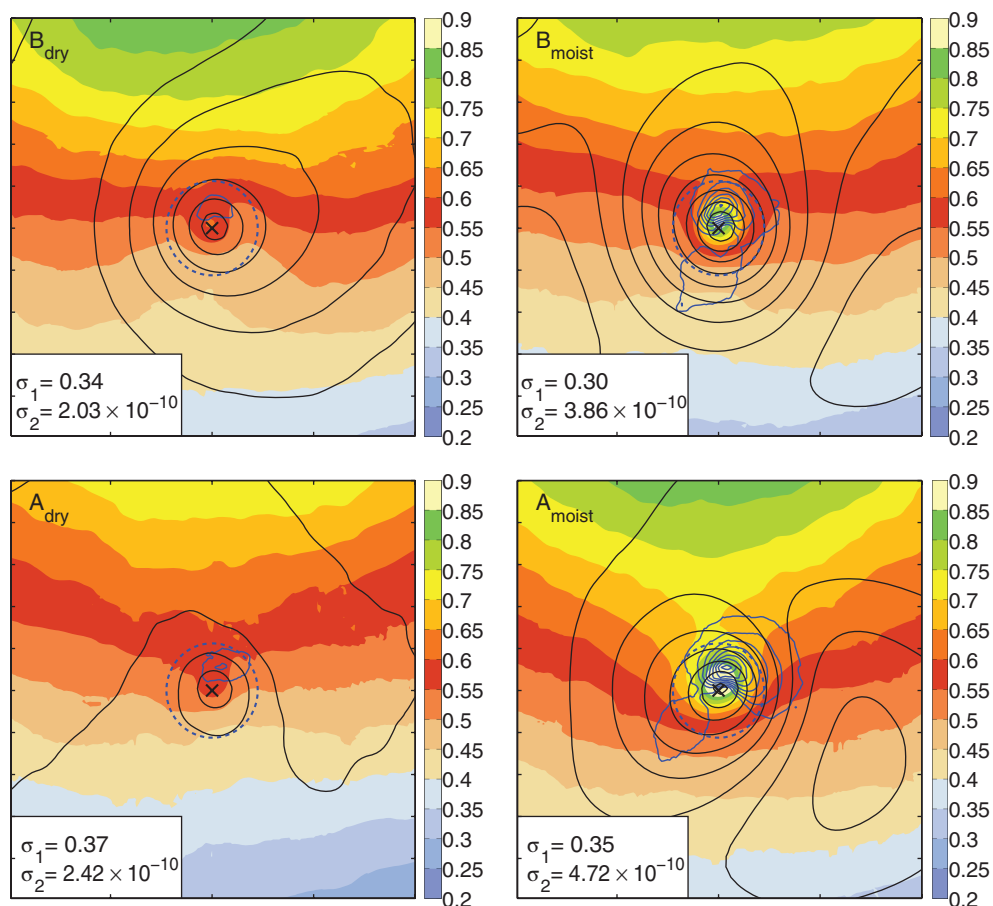


Figure 8. PV_{low} (in colour), $FGEN_{850}$ (solid blue contour) and sea-level pressure (thin solid lines) as in Figure 6(b), but for the edge classes A_{dry} , A_{moist} , B_{dry} and B_{moist} .

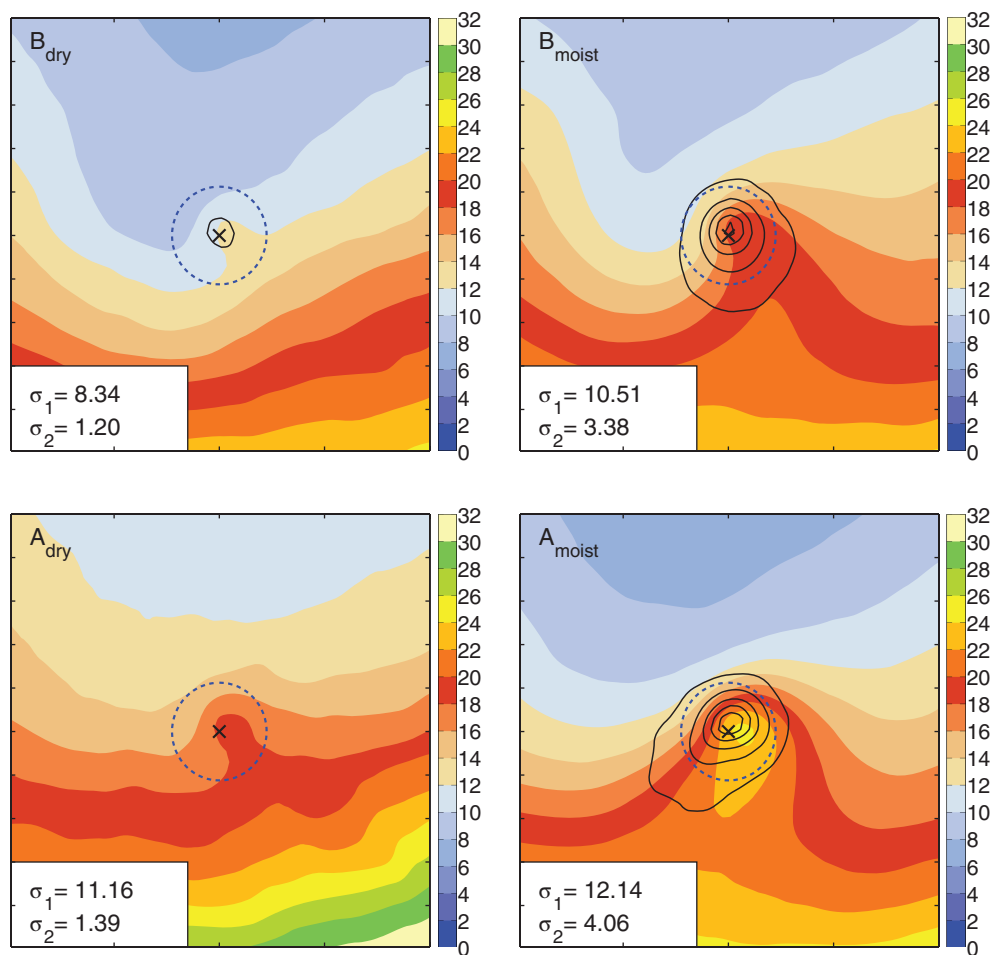


Figure 9. $Qint$ (in colour) and RR_{gh} (contours) as in in Figure 6(c), but for the edge classes A_{dry} , A_{moist} , B_{dry} and B_{moist} .

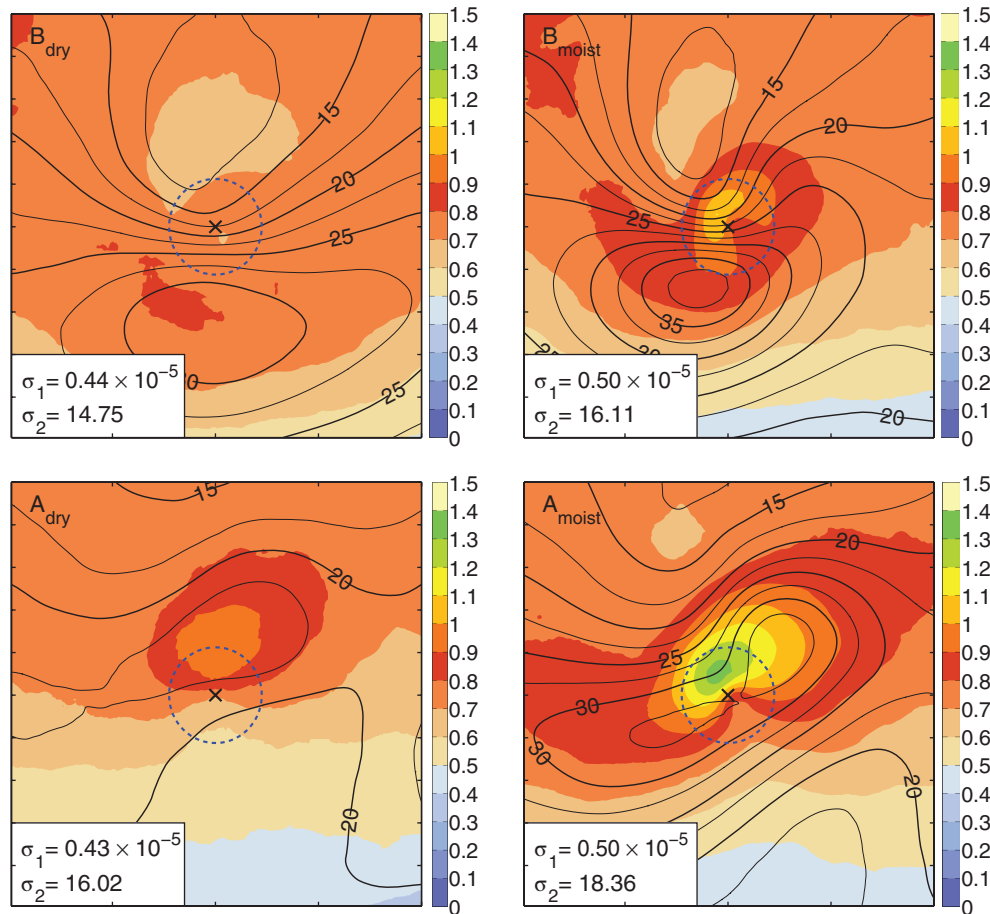


Figure 10. EADY (in colour) and VELJET (contours) as in Figure 6(d), but for the edge classes A_{dry} , A_{moist} , B_{dry} and B_{moist} .

The strongest PV_{up} signal is found for class B_{moist} . Cyclogenesis develops on the downstream side of a pronounced, rather narrow PV trough, exhibiting strong PV gradients at cyclogenesis. The strong PV signal is associated with a correspondingly strong signal in $qg\omega_{top}$. Indeed, the maximum amplitude of this field, nearly perfectly located at cyclogenesis, is several times larger than for all the other classes. As for all classes, the ascending branch of the $qg\omega_{top}$ dipole is considerably stronger than the descending branch on the upstream side of the narrow PV trough. From Figure 4, we expect this class to have similar tropospheric stabilities as class B_{dry} .

Finally, we discuss class A_{moist} , where cyclogenesis is again located downstream of a positive PV_{up} anomaly. As for A_{dry} , the tilt between the surface cyclone and the PV_{up} signal is fairly large. A pronounced ridge prevails east of the cyclogenesis location. With respect to $qg\omega_{top}$, this class is associated with weaker forcing for ascent than class B_{moist} , but much stronger than for A_{dry} .

4.2.2. Low-level and moist processes

Figure 8 shows low tropospheric structures (see Figure 6(b) for class M). For class A_{dry} , only a very weak positive PV_{low} anomaly is superimposed on a background meridional PV gradient. Similarly, $FGEN_{850}$ also exhibits a weak signal northeast of A_{dry} cyclogenesis. Both signals indicate only weak low-level processes for these genesis events, which is in line with Figure 4. Class B_{dry} is very similar to A_{dry} , with very weak signals of PV_{low} and $FGEN_{850}$. The only noteworthy differences are an enhanced background PV gradient for B_{dry} and the presence of several closed SLP contours in the composite.

For class A_{moist} , the situation changes significantly. A rather strong and small-scale cyclonic PV_{low} anomaly is present very close to the cyclogenesis location, indicating that diabatic processes play an important role. Furthermore, this class shows the highest values of $FGEN_{850}$, with a maximum slightly north of

cyclogenesis. A second, although weaker, maximum of $FGEN_{850}$ is located to the southwest. The overall pattern is reminiscent of a warm and cold front, respectively, suggesting that cyclogenesis occurs near a local minimum of a generally strongly frontogenetic region. The SLP pattern shows a dipole with a surface anticyclone to the east of the genesis location. This points to the potential role of warm-air advection, in line with the biplot in Figure 4, where T_{ADV850} points to the quadrant of A_{moist} . For class B_{moist} , the low-level fields are very similar. However, here the PV_{low} signal almost coincides with the PV_{up} anomaly (Figure 7), leading to a particularly strong vertical coupling of the two PV anomalies. The SLP composite of B_{moist} has the strongest cyclone and a weaker downstream anticyclone than in A_{moist} .

From Figure 8, it is evident that PV_{low} differs significantly between the dry and moist classes, due to the differing degrees of latent heat release in clouds. This is confirmed by the fields shown in Figure 9. As expected, Q_{int} shows a strong southward gradient for all classes. A_{dry} shows a weak positive disturbance of Q_{int} at the cyclogenesis location and almost zero values of RR_{6h} . Class B_{dry} is very similar to class A_{dry} in terms of Q_{int} and RR_{6h} , with generally lower values of Q_{int} , due to the more poleward cyclogenesis positions (Figure 5). The slightly stronger signal of RR_{6h} is consistent with a stronger QG forcing for ascent in B_{dry} than in A_{dry} (Figure 7).

From Figure 4, we expect similar values of RR_{6h} for classes A_{moist} and B_{moist} , but differences in Q_{int} , because the Q_{int} vector points into the A_{moist} class and away from the B_{moist} class. Indeed, the composites in Figure 9 reveal strongly positive anomalies of Q_{int} and RR_{6h} at cyclogenesis. For class B_{moist} , cyclogenesis occurs at the northern tip of the high- Q_{int} tongue (similar to the two dry categories). A distinct difference of the two moist classes is the orientation of the Q_{int} tongue and the clear local maximum of Q_{int} to the southeast of A_{moist} cyclogenesis. Furthermore, the absolute values of Q_{int} are higher for A_{moist} than for B_{moist} .

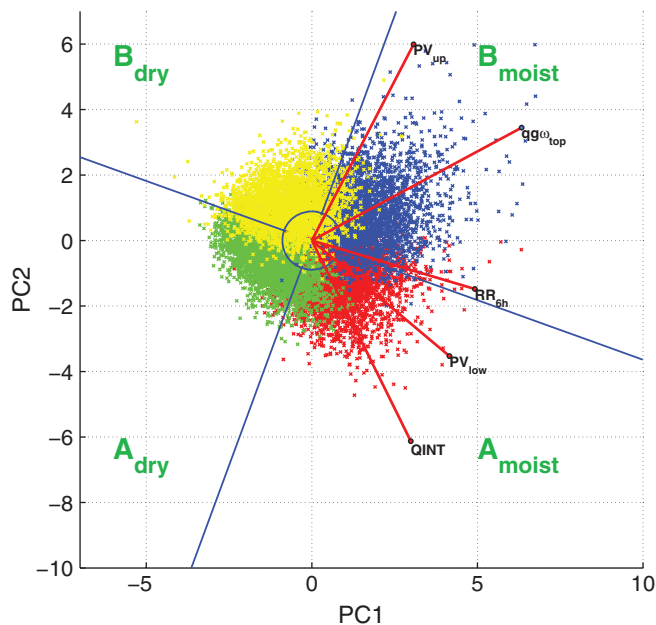


Figure 11. Biplot with reduced number of precursors. The red lines represent the coefficients of the precursors. If they point towards positive (negative) anomalies, they have a red (blue) dot at the end. They are labelled with the abbreviations listed in Table 1. The crosses determine the scores and their colours represent the original classification (yellow = B_{dry} , blue = B_{moist} , green = A_{dry} , red = A_{moist} , grey = M). The blue solid lines define the class borders of the classification with a reduced number of precursors.

4.2.3. Measures of baroclinicity

Finally, Figure 10 shows two measures of baroclinicity: the Eady growth rate $EADY$ and upper-level jet velocity $VELJET$. Class A_{dry} exhibits moderate values of $EADY$, with a weak but distinct maximum north of the cyclogenesis location. $VELJET$ has fairly uniform and moderate values ($\sim 20 \text{ m s}^{-1}$). The absence of a well-defined jet streak is consistent with the weak values of $qg\omega_{\text{top}}$ (Figure 7). B_{dry} exhibits the lowest values of $EADY$ of all classes. $VELJET$, however, shows a strong meridional gradient and a distinct maximum of 30 m s^{-1} relatively far south of B_{dry} cyclogenesis.

The situation clearly changes for the moist classes. A_{moist} shows the highest values of $EADY$ just northwest of the cyclogenesis location. The region of enhanced values of $EADY$ is elongated from southwest to northeast. In the same region, high values of $VELJET$ occur. Therefore cyclogenesis in class A_{moist} takes place just beneath the jet streak centre. Values of $EADY$ are a bit weaker for B_{moist} , but the maximum is again located to the northwest. As before, the $VELJET$ signal is interesting, mainly because of the very high values ($> 35 \text{ m s}^{-1}$) southwest of cyclogenesis, indicating that cyclogenesis occurs in the left exit of a jet streak. This is in line with the very strong upper-level forcing found for this class (Figure 7).

5. Discussion

In this section, we discuss several aspects of our method and results. First, the question is addressed as to the robustness of the classification with respect to the number of precursors. Then the identified classes of cyclogenesis are compared qualitatively with the classification scheme by Petterssen and Smebye (1971), before we discuss the genesis classification for a selection of well-known cyclones.

5.1. Reproducibility of classification with few precursors

The sensitivity of the classification has been tested with a strongly reduced set of precursors. To this end, only very important precursors (according to the biplot in Figure 4), i.e. $qg\omega_{\text{top}}$, PV_{up} ,

PV_{low} , RR_{6h} and $Qint$, were used. Figure 11 reveals a very similar pattern to the original biplot; however, with the vectors rotated by 20° clockwise. This rotation might occur because, in this lower-dimensional phase space, PC1 and PC2 are associated with a similar explained variance. In order to compare the classes of the full dataset with the reduced one, the class borders are also rotated. With this rotation, 70% of the events are classified in the same way as with the full set of precursors. As can be noticed in Figure 11, the highest density of differently classified events occurs close to the class borders. This indicates that the characteristics of the first two principal components and of the classification are fairly robust, i.e. they can essentially be reproduced even with a strongly reduced set of precursors. In other words, this brief sensitivity study indicates that considering many precursors, where some of them eventually turn out to be of low importance, does not impede a meaningful classification.

We also note that, with the reduced set of precursors, the explained variance of PC1 and PC2 amounts to 73.8%, which is considerably higher than in the original classification with 41.2%. This can be explained by the lower dimensionality of the phase space, such that more variance must be represented by the first two PCs.

5.2. Comparison with previous classifications

First, the question arises as to how this classification compares with the scheme by Petterssen and Smebye (1971). It is important to note that their classification also considers the temporal evolution of cyclones, in contrast to our approach, which focuses on flow structures at the time of genesis only. Since Petterssen and Smebye (1971) do not refer explicitly to moist dynamical processes, their classification focuses more on our PC2. Instead of two distinct categories, we find a rather gradual transition from cyclones that develop with a weak upper-level trough (similar to type A) to those that develop downstream of a pronounced upper-level trough (similar to type B). Cyclogenesis with very weak upper-level forcing (type A) is very rare and occurs in extreme cases of class A_{moist} associated with strongly negative values of PC2 (e.g., cyclone ‘Kyrill’; see below). In agreement with Petterssen and Smebye (1971), for these events, baroclinicity (e.g. $EADY$) is particularly strong. Moreover, cyclogenesis with negative values of PC1 occurs in an area where the upper-level QG forcing is weak. The later classification by Deveson *et al.* (2002) included a third type, C, for which moist processes are also essential. Type C cyclones also have a pronounced upper-level forcing and therefore are likely similar to our events in class B_{moist} . In summary, the existing ABC categorization is not unlike the more extreme cases in our classes A_{moist} (type A), B_{dry} (type B) and B_{moist} (type C). However, a very important result of this study is that, if considering a high-dimensional precursor phase space, cyclogenesis events form a continuum and not a set of distinct types. This continuum has such a large spread that the four outer classes nevertheless have strongly differing and physically distinguishable characteristics.

5.3. Classification of selected cyclogenesis events

Figure 12 shows the PC1 and PC2 scores of some subjectively selected and generally well-known cyclone events. They are listed in Table 2 and, where available, references are given to studies describing these events. Most of these genesis events are associated with enhanced values of PC1, i.e. intense low-level moist processes. Exceptions are the cases ‘Dry’ (B_{dry}) and ‘Algeria’ (A_{dry}), with negative values of PC1. The conditions during cyclogenesis, investigated in this study, cannot be directly attributed to the future cyclone evolution; a detailed investigation of the linkage between genesis conditions and the subsequent life-cycle would be an interesting route for future research. The complexity of this linkage is illustrated by the selected events: ‘Algeria’, a cyclone associated with heavy precipitation

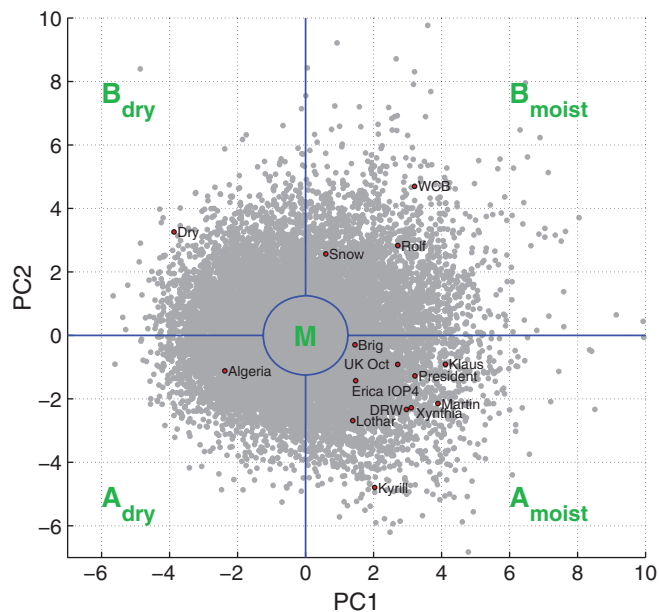


Figure 12. Biplot with selected cyclogenesis cases listed in Table 2 (red dots). The grey dots represent the scores of all cyclogenesis events. The blue solid lines define the class borders of the classification (as in Figure 4).

and flooding in the area of Algiers, appears in class A_{dry} , whereas most other cyclones with high impact are in one of the moist classes (e.g. ‘Brig’, related to a damaging flood event in the Southern Alps, ‘Rolf’, which caused heavy precipitation in the Mediterranean, ‘Lothar’, producing strongly damaging winds in Central Europe, and ‘Snow’, associated with high amounts of wet snow over northern Germany).

6. Conclusions

This study introduced a novel approach to classify events of extratropical cyclogenesis. Compared with previous approaches, which were based on the physical understanding of the key precursor fields, our approach is more objective in the sense that it includes a large set of potential precursors and then determines statistically those that show the largest variability across a large ensemble of 16 029 genesis events. Our analysis indicates that no obvious cyclone clustering occurs within the 30-dimensional precursor phase space. However, a projection of all 16 029 cyclogenesis events on the first two PCs of the precursor phase space allows for a meaningful separation of cyclones into five classes. The first two PCs (PC1 and PC2)

represent upper-level forcings and lower-level diabatic processes, respectively. Moist processes (including diabatically produced low-level PV) and upper-level PV are almost uncorrelated and characterize the first two principle components of the precursor phase space. Baroclinicity is correlated with moist processes, but less variable. The QG upper-level forcing for ascent correlates equally with upper-level PV and baroclinicity, which agrees nicely with the theoretical understanding that baroclinicity and flow changes along isentropes determine the amplitude of the QG forcing for ascent. It is interesting that our objective approach identifies upper-level PV and moist processes as the most variable precursors of extratropical cyclogenesis, corroborating results from many case studies and climatological analyses, which highlighted the key role of these dry and moist parameters.

We divided the phase space into five equally populated classes and termed the outer classes as A_{dry}/A_{moist} and B_{dry}/B_{moist} . The AB terminology is a reference to the classical Petterssen and Smebye (1971) classification, with comparatively weak (A) versus strong (B) upper-level flow anomalies near cyclogenesis. Dry and moist indicate that some genesis events occur in fairly dry regions with weak/no precipitation and diabatic low-level PV production and others occur in a region where vertically integrated moisture has a local maximum and latent heating associated with intense precipitation leads to a pronounced cyclonic PV anomaly at low levels. Composites of the genesis environment for the five classes shows that they can be characterized as follows.

- A_{dry} : cyclogenesis occurs in a region with very weak QG upper-level forcing for ascent beneath a shallow upper-level ridge, moderate baroclinicity, moderate humidity but no precipitation.
- A_{moist} : cyclogenesis occurs downstream of a large-scale upper-level trough with moderate QG upper-level forcing at the axis of an upper-level jet and at the poleward edge of a very moist low-level airmass. Baroclinicity is very strong, as are precipitation and the amplitude of the cyclonic low-level PV anomaly.
- B_{dry} : cyclogenesis occurs downstream of an upper-level cut-off-like flow situation with moderate QG upper-level forcing in a typically dry environment and weak baroclinicity.
- B_{moist} : cyclogenesis occurs at the downstream flank of a pronounced and comparatively narrow upper-level trough with very strong QG forcing for ascent at the left exit of an intense jet streak. Humidity is moderate, precipitation and low-level diabatic PV production are strong.

Table 2. Table with a selection of extratropical cyclogenesis cases.

Name	Abbrev.	Paper	Lat/Lon	t_{gen}	Class	p_{gen}	p_{min}
President's day storm	President	Whitaker <i>et al.</i> (1988)	33°N 78°W	1979/02/19 0000 UTC	A_{moist}	1018.4	979.6
UK October storm	UK Oct	Hoskins and Berrisford (1988)	32°N 72°W	1987/10/12 1200 UTC	A_{moist}	1008.6	958.7
Erica IOP4	Erica IOP4	Neiman and Shapiro (1993)	36°N 75°W	1989/01/01 1800 UTC	A_{moist}	1013.3	967.5
Brig flood	Brig	Massacand <i>et al.</i> (1998)	37°N 01°E	1993/09/22 0600 UTC	A_{moist}	1011.0	1001.1
Lothar	Lothar	Wernli <i>et al.</i> (2002)	37°N 63°W	1999/12/24 0600 UTC	A_{moist}	1010.9	976.5
Martin	Martin	-	46°N 24°W	1999/12/27 0000 UTC	A_{moist}	1000.5	969.8
Algeria	Algeria	Tripoli <i>et al.</i> (2005)	35°N 01°E	2001/11/05 1200 UTC	A_{dry}	1018.6	1018.1
Dry event	Dry	-	66°N 09°E	2002/10/20 0600 UTC	B_{dry}	1002.7	1002.7
Snow event	Snow	Frick and Wernli (2012)	59°N 10°E	2005/11/24 1800 UTC	B_{moist}	989.8	974.1
DRW	DRW	Boettcher and Wernli (2011)	32°N 78°W	2005/12/18 1200 UTC	A_{moist}	1017.7	975.8
Kyrill	Kyrill	Fink <i>et al.</i> (2009)	44°N 61°W	2007/01/16 0600 UTC	A_{moist}	1002.6	964.0
Klaus	Klaus	Liberato <i>et al.</i> (2011)	44°N 29°W	2009/01/23 0600 UTC	A_{moist}	994.2	967.5
WCB	WCB	Joos and Wernli (2012)	45°N 23°W	2009/01/29 0600 UTC	B_{moist}	987.7	972.9
Xynthia	Xynthia	Ludwig <i>et al.</i> (2014)	26°N 41°W	2010/02/24 1800 UTC	A_{moist}	1012.7	969.0
Rolf	Rolf	-	41°N 00°E	2011/11/04 1800 UTC	B_{moist}	995.3	995.3

The column **Name** contains the name or a short description of the event and the column **Abbrev.** the abbreviation that is used in Figure 12. **Paper** shows one selected reference for this case. **Lat/Lon** denotes the geographical location of the cyclogenesis and t_{gen} its date and time. **Class** shows the assigned class. p_{gen} indicates SLP in the cyclone centre during genesis and p_{min} the minimum SLP along the whole track.

- *M*: the fifth class represents genesis events with average characteristics. For events in this class, none of the main precursors is particularly strong or weak.

Most of the well-known and high-impact cyclones considered in this study (the US President's Day cyclone, ERICA IOP4, the UK October storm and the European wind storms *Lothar*, *Klaus* and *Xynthia*) belong to class *A_{moist}*, i.e. the class with, on average, the highest values of baroclinicity and vertically integrated water vapour. Note that our classification only considered cyclones at the time of genesis and an analysis of how the evolution of cyclones projects on our parameter phase space would be a potentially attractive extension of this study. Another valuable extension could be to investigate cyclogenesis in the Southern Hemisphere also and in different seasons.

Finally, we comment on the limitations of our approach and the resulting classification. The limitations mainly concern (i) the accuracy of the dataset, (ii) sensitivity to the cyclone-tracking scheme, (iii) the choice of precursor parameters and (iv) the classification. Our results are based on ERA-Interim, one of the widely used state-of-the-art reanalysis datasets. Still, potential small-scale flow structures are not captured by this dataset (because its resolution is too coarse) and therefore cannot be identified as a potential precursor. We defined 'genesis' as the first point of a cyclone track and therefore our set of genesis events depends on the cyclone identification and tracking scheme utilized. The tracking intercomparison study by Neu *et al.* (2013) showed that most cyclone-tracking schemes agree fairly well for intense cyclones during the central period of their life cycle. However, the early and late phases of a cyclone, i.e. the phases near genesis and lysis, are more sensitive and depend on the tracking technique adopted. Therefore, an alternative tracking scheme (e.g. a scheme that tracks low-level relative vorticity maxima) might cause cyclones to have a slightly different time and location of cyclogenesis. It might be an interesting task for future research to test the sensitivity of the genesis conditions identified to the choice of tracking scheme – however, given that our main genesis regions are in very good agreement with other cyclone climatologies, we are convinced that our genesis locations are meaningful for the classification exercise adopted. A brief sensitivity test presented in section 5 revealed that omitting some of the less important precursors would not change the results of the present study substantially. More difficult to assess is whether or not we missed any important precursors in our analysis. This cannot be excluded, but given the large set of precursors and the thorough literature search, we regard this as unlikely. Finally, there is also some subjectivity in our classification. The number of classes is somehow arbitrary, but five appears to be a sensible choice, given that the composites reveal strongly differing structures. The way the classes were constructed (division of the PC1–PC2 phase space) in more or less equal segments is a pragmatic and somehow objective approach, given that there are no well-defined a priori clusters in the phase space. It is important to be aware of the limitations of and specific choices made in our approach, which nevertheless provides an important step forward in understanding the main factors that determine the large variability of cyclogenesis in the Northern Hemisphere extratropics.

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Supporting information

The following supporting information is available as part of the online article:

File S1. Detailed description and composites of precursors.

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